

IP addresses: 1

Advanced: Practice 1  

Select the **two** statements about IP addresses which are correct.

- ☐ IP addresses are assigned to a device by the manufacturer and can't easily be changed
- ☐ Every device connected to the internet has an IP address
- ☐ Mobile phones do not have IP addresses
- ☐ An example of an IP address is **192.168.4.23**

Quiz:

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Application layer protocols

Core: Practice 2  

LLM marked question

A protocol is a set of rules that determines how two devices will communicate. Most network communications rely on sets of protocols that are organised in layers.

Name one network protocol that resides on the application layer of the TCP/IP model and briefly describe what it is used for.

[2 marks]

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Von Neumann architecture 3

Advanced: Practice 1  

John von Neumann proposed a general purpose computer that is sometimes referred to as the "three box model". Which of the following statements provides a correct definition of a von Neumann machine?

- ☐ A processor, clock, and main memory linked by a system bus
- ☐ A processor, main memory, and secondary storage linked by a system bus
- ☐ A processor, cache, and main memory linked by a system bus
- ☐ A processor, main memory, and I/O controllers linked by a system bus

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Special purpose registers

Core: Practice 2  

Which sentence below best describes dedicated (special-purpose) registers?

- ☐ Registers that are used to store the results of intermediate calculations.
- ☐ Registers that is defined in software and can be customised by the user.
- ☐ Registers that have a specific role in the fetch-execute cycle.
- ☐ Registers that hold memory addresses.

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Complete truth table for logic circuit 2

Core: Practice 2

Study the logic circuit shown in **Figure 1** and the truth table below.

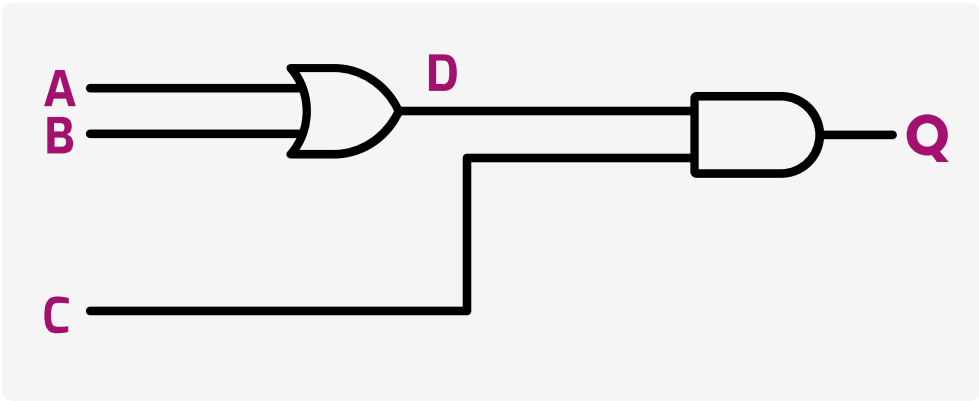


Figure 1: A circuit diagram

Complete the truth table for the logic circuit diagram in **Figure 1**. The column **D** represents the output of the logic gate with inputs A and B. The final output is in column **Q**.

A	B	C	D	Q
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

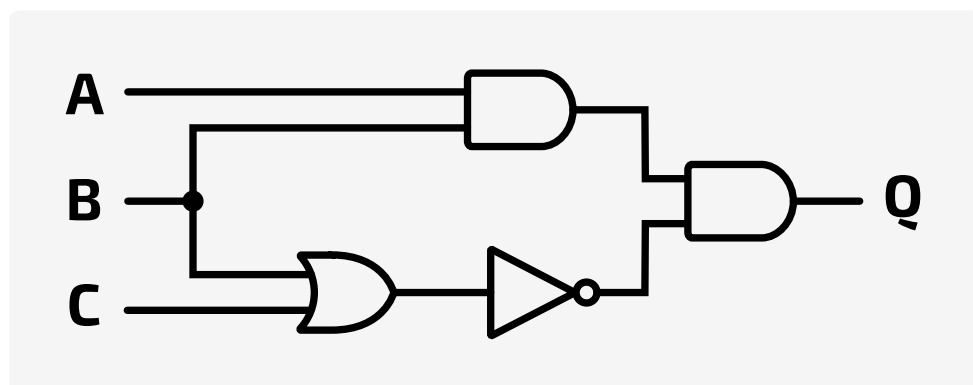
Quiz:

Logic circuit for expression

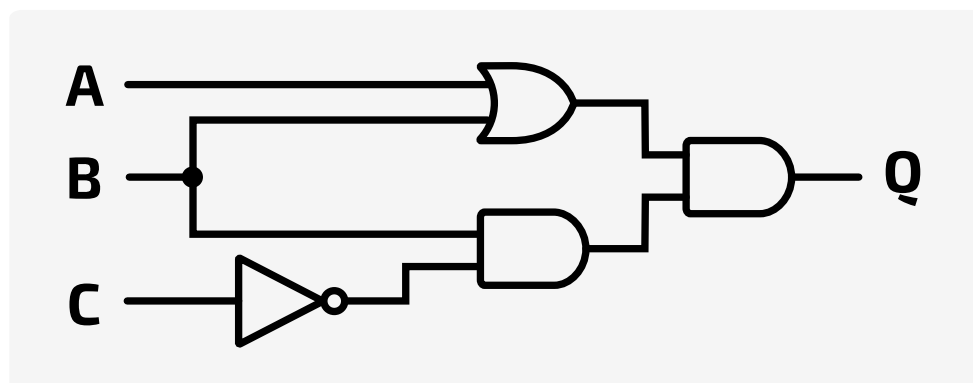
Core: Challenge 2 ■■

Which of the circuits below represents the following Boolean expression?

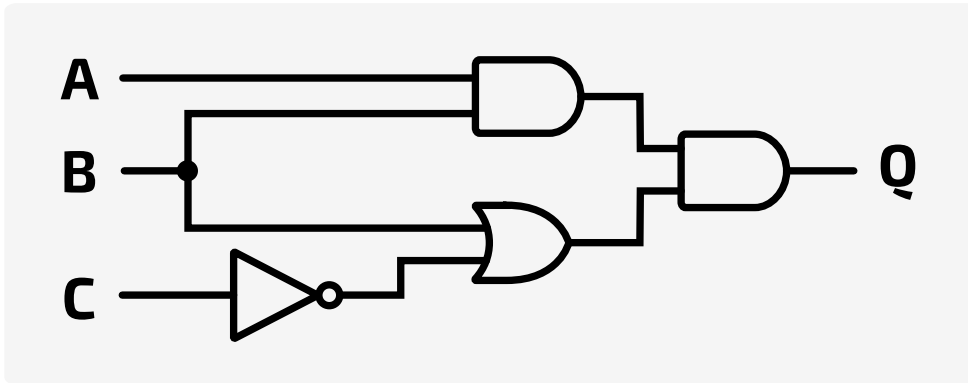
$$Q = (A \wedge B) \wedge (\neg C \vee B)$$



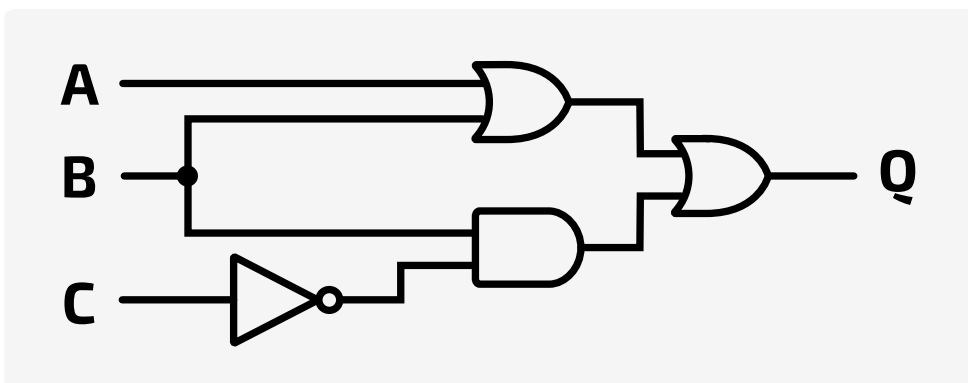
Option A



Option B



Option C



Option D

- ☐ Option A
- ☐ Option B
- ☐ Option C
- ☐ Option D

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Expression for problem 2

Advanced: Challenge 1



A library system needs to indicate if a user is allowed to borrow a book. The decision is based on several criteria.

- No one can borrow a book if they have an unpaid fine on their account.
- The maximum number of books a user can borrow is six unless they are an A level student or they have been waiting for the book for more than four weeks.

The following inputs represent the listed conditions:

- F - True if there is an unpaid fine (F) on the account.
- L - True if the account has six or more books on loan (L).
- A - True if the account belongs to an A level (A) student.
- W - True if the book was requested more than 4 weeks (W) ago.

The output B will be True if the student is allowed to borrow the book (B).

Choose the correct Boolean expression to match the logic of the problem statement.

- ☐ $B = \neg F \wedge (W \vee A \vee L)$
- ☐ $B = \neg F \vee (W \wedge A \wedge \neg L)$
- ☐ $B = \neg F \wedge (W \vee A \vee \neg L)$
- ☐ $B = F \wedge (W \vee A \vee L)$
- ☐ $B = \neg F \vee (W \vee A \vee \neg L)$
- ☐ $B = F \wedge (W \vee A \vee \neg L)$

Quiz:

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Two's complement: binary to denary 2

Advanced: Practice 1  

Convert the binary number 00101011 to denary. The binary value is encoded as an 8-bit **two's complement** number.

Type your answer as a **signed decimal number** (e.g. +3.75). Do not leave any spaces in your answer.

Quiz:

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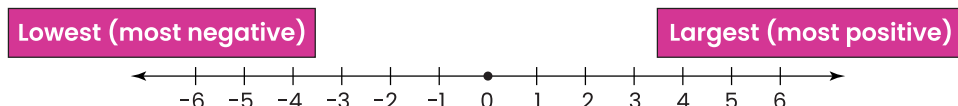


Two's complement: lowest to largest

Advanced: Practice 2  

The integers shown below are represented using **two's complement**.

The lowest number is defined as the **most negative number** (i.e. the negative number that is furthest away from 0), and the largest number is defined as the **most positive number** (i.e. the positive number that is furthest away from 0).



A number line

Put the following two's complement numbers into order from **lowest (most negative)** to **largest (most positive)**. The lowest number should appear at the top of the list and the largest number at the bottom of the list.

Available items

11001110

01111111

11111111

01001110

10000000

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Two's complement: range 1

Advanced: Practice 1  

Signed integers can be stored in two's complement form. What is the range of values that can be stored using 8 bits in two's complement?

- ☐ +127 to -128
- ☐ +256 to -256
- ☐ +255 to -256
- ☐ +128 to -128

Quiz:

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